

A 16 MIB COMPUTER

**Our AI agent was  
using 4% of its  
machine.**

We gave it 59%. On purpose, and it made the thing  
genuinely better.

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A vector database with no database under it —  
and no operating system either.

## THE MACHINE IN QUESTION

# One program. No OS beneath it.

A Telegram agent compiled as a **BareMetal unikernel**: no kernel, no shell, no init, no container, no disk. A 2.9 MB image that boots straight into **main()**.

Its memory ceiling is not a plan you upgrade. It is the platform maximum:

```
{ "error": "ramMib exceeds the maximum of 16" }
```

## THE PROBLEM WORTH SOLVING

# It could only find what you filed.

The agent remembers things you tell it. But recall was exact-key lookup — ask "what do you know about boot times?" and it found nothing, because you had filed it as "the unikernel starts in 31 ms".

The notes were there. The words were wrong.

## WHAT WE BUILT

**Every memory, embedded,  
resident in RAM.**

Each note is embedded once into a **768-dimension vector** and kept in a static array. A question is embedded too, then compared against every resident vector.

One pass over an array. **No tree. No index structure. No database. No disk** — this machine has none of those things and now needs none of them.

THE BUDGET, AGAINST A HARD 16

# 9.37 MiB, claimed at link time.

vectors       $2560 \times 768 \times 4 \text{ B}$       = 7.50 MiB

text           $2560 \times 192 \text{ B}$               = 0.47 MiB

chat ids       $2560 \times 8 \text{ B}$                  = 0.02 MiB

buffers      arena/HTTP/JSON      = 1.38 MiB

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total                                      = 9.37 MiB of 16

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2,560 memories, searchable by meaning, on a machine with 16 MiB and no OS.

THE PART THAT SURPRISED US

8 MiB of RAM cost 31 KB of image.

2,880,384 bytes uploaded before

2,911,872 bytes uploaded after

Uninitialised statics aren't stored in the file. The whole 31 KB difference is code — the machine simply wakes up with the room already claimed.

## GOTCHA #1 — THE SILENT ONE

# Truncated embeddings are not unit vectors.

The model returns 3072 dimensions and truncates to 768 on request. We assumed the result was still unit length.

It isn't. Measured norm: 0.59.

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Cosine over unnormalised vectors ranks LONGER notes above CLOSER ones. Silently. Plausibly. You would never catch it from the output.

## GOTCHA #2 — THE BORING ONE THAT MATTERS

# We parse 3,072 floats by hand.

There is no heap in this program. JSON goes through a fixed bump arena, and a batch of four embeddings would become 3,072 separate nodes in it.

So the embeddings are scanned out of the response with `strtod`. It costs nothing, and the shape is safe to assume — this program wrote the request.

THE DECISION WE ARE PROUDEST OF

**1 in 4000**

**results on this platform  
come back wrong.**

We had measured it that morning: this unikernel corrupts 64-bit arithmetic under load. A brute-force dot product over a big array is *exactly* that loop shape.

SO WE BUILT IT ANYWAY

# Because of how it fails, not whether.

A corrupted score reorders two neighbours in a ranked list. It cannot crash. It cannot invent a memory that was never stored.

A feature needing exact arithmetic would have been the wrong thing to put on this machine. **The constraint picked the feature** — not the other way round.

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Know your platform's failure mode, then choose work that degrades instead of breaking.

MEASURE FIRST, THEN CLAIM

## 13 MiB boots. 14 does not.

Before writing a line of the feature, we probed the real ceiling: allocate X MiB, write every page, read it all back, see whether the machine survives.

Then sized the index to fit with margin, rather than guessing and finding out in production.

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5 of 5 test queries return the note a human would pick. Search time over the index: 4.2 microseconds.

IT IS LIVE RIGHT NOW

# Ask it something. It will remember.

```
[+] agent up. ram=16 MiB static=9588 KB (59% of RAM)
[+] semantic index: 2560 slots × 768 dims
    = 8180 KB resident, searched in RAM
```

Open source, C, no heap, no OS.

```
github.com/tabibazar/baremetal-agent
```

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Built on Return Infinity's BareMetal. The arithmetic bug is reported, with a 60-line reproducer.